

When do sequence-only transformers beat boosted trees for CRISPR-Cas9 on-target activity prediction? A head-to-head across mammalian and bacterial corpora.

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Abstract

On-target activity prediction for CRISPR-Cas9 single-guide RNAs (sgRNAs) is dominated in published practice by gradient-boosted trees over hand-engineered sequence features (Rule Set 2, Doench et al. 2016) for mammalian systems and by CNN+RNN architectures (crisprHAL, Ham et al. 2023) for bacterial systems. The competitiveness of small sequence-only transformers, which are increasingly the default in genomic modelling at scale, has not been carefully measured against these specific baselines at corpus sizes typical for CRISPR datasets. In this work, we reproduce the Rule Set 2 baseline on the Doench 2016 mammalian corpus (5,310 sgRNAs) and the published crisprHAL test set on the WT-SpCas9 bacterial corpus (33,743 sgRNAs), and train a small encoder-only transformer ($\approx 200,000$ parameters) against each. On the mammalian corpus we observe statistical ties (Spearman ρ 0.420 [0.343, 0.488] for the transformer vs. 0.459 [0.389, 0.527] for the baseline; AUC identical to three decimals). On the bacterial corpus the transformer is cleanly outperformed by the boosted-tree baseline (0.544 [0.526, 0.562] vs. 0.626 [0.608, 0.642]; non-overlapping bootstrap CIs). A cross-kingdom transfer-learning experiment, in which the mammalian-trained transformer is fine-tuned on bacterial data, did not improve over fresh initialisation. We discuss three plausible drivers for the contrast — cleaner bacterial signal, the cost of cropping the published 378-nt bacterial window to a 43-nt input, and the architectural design choices the crisprHAL authors made — and outline future work to disambiguate them.

Keywords: CRISPR-Cas9, sgRNA, on-target prediction, gradient boosting, transformers, bacterial genome editing.

1. Introduction

CRISPR-Cas9 is the dominant programmable genome-editing system used in mammalian cell engineering, agricultural genomics, and industrial-strain construction. Cleavage efficiency of a CRISPR-Cas9 ribonucleoprotein at a chosen genomic site depends on the 20-nucleotide spacer sequence of the sgRNA and the flanking sequence context [1]. Predicting on-target activity from sequence alone — without expensive empirical screening — is a core problem in CRISPR design, with thousands of citations on published predictors and several mature web tools used in production laboratories.

Two prediction systems dominate published practice. For mammalian cells, **Rule Set 2** [1] — a gradient-boosted regressor over hand-engineered sequence features (position-specific mono- and dinucleotide one-hots, position-independent counts, melting-temperature thermodynamics, and a TTTT terminator flag) — remains the de facto baseline; its training corpus of 5,310 measured sgRNAs across 17 human and mouse genes is the canonical “small but clean” CRISPR-Cas9 benchmark. For bacterial systems, **crisprHAL** [2] is a recently-published CNN+RNN architecture trained on $\approx 33,000$ sgRNAs targeting an *E. coli* plasmid in a two-plasmid positive-selection assay specifically designed to separate cleavage activity from cellular toxicity.

Sequence-only transformer architectures, which now dominate at the scale of whole-genome modelling (DNABERT, Nucleotide Transformer), have been less carefully measured against these specific CRISPR baselines at corpus sizes typical of CRISPR-Cas9 efficiency screens (5–30 k labeled sgRNAs). The published literature on deep-learning sgRNA predictors uses heterogeneous test splits and a variety of architectures, making it difficult to ask the methodological question we focus on here: **for a fixed corpus and a fixed evaluation protocol, does a small sequence-only transformer match, beat, or lose to a strong hand-engineered baseline?**

We measure this directly on two corpora, with identical bootstrap methodology and shared evaluation code, and report the result honestly in both directions. We then probe whether cross-kingdom transfer-learning — using a mammalian-trained transformer to initialise a bacterial fine-tune — helps in the bacterial regime where small transformers most need help.

1.1 Contributions

A reproduced Rule Set 2 baseline at Spearman ρ 0.459 [0.389, 0.527] on a leave-genes-out test split of the Doench 2016 corpus, in agreement with the paper’s reported ~ 0.55 within the bootstrap CI.

A head-to-head, bootstrap-CI’d comparison of a ≈ 200 k-parameter encoder-only transformer against the Rule Set 2 baseline on Doench 2016, finding statistical equivalence.

A three-way comparison on the crisprHAL WT-SpCas9 bacterial corpus between (i) an xgboost regressor over Rule Set 2-style features adapted to the bacterial 43-mer window, (ii) a fresh transformer of identical architecture trained from random initialisation, and (iii) the same transformer fine-tuned from the mammalian Phase-2 checkpoint.

An analysis of why the mammalian-corpus statistical tie does not survive transfer to the bacterial corpus, and the implications for architecture selection in CRISPR design tooling.

2. Methods

2.1 Datasets

Doench 2016 mammalian corpus [1]. We use the published FlowCytometry+Resistance (“V3”) dataset distributed in the Microsoft Research Azimuth repository, comprising 5,310 sgRNA target-site sequences across 17 human and mouse genes, each measured under up to four drug-selection conditions. Each

row is a 30-nucleotide context window (4 nt 5' flank, 20 nt protospacer, 3 nt NGG PAM, 3 nt 3' flank). Activity is encoded as a within-gene-within-drug rank in [0, 1].

crisprHAL WT-SpCas9 bacterial corpus [2]. We use the WT-SpCas9 training and test CSVs from the maintained crisprHAL repository, comprising 33,743 sgRNAs (26,994 train + 6,749 test) measured in an *E. coli* two-plasmid positive-selection assay. Each row is a 378-nt context window with the labeled target's PAM always at positions 209–211 (verified across all rows). Activity is a log-enrichment score from Illumina sequencing of pre- and post-selection sgRNA pools, range approximately [−9, +6]; the corpus is not rank-normalised at distribution time.

2.2 Splits

For Doench 2016 we use **leave-genes-out** splits to avoid within-gene correlation contamination, the protocol the original paper uses for its reported numbers. The 17 unique target genes are assigned to train/val/test via greedy longest-processing-time bin-packing against fractional targets of 80/10/10, with seed=42. This yields a 4,209-row train, 551-row val, and 550-row test split with no gene overlap. The MED12 gene alone contains 1,848 rows (≈35% of the corpus); naïve random gene-permutation produces a 40% val split, hence the bin-packing.

For the crisprHAL bacterial corpus the published train/test partition is preserved; we hold out a deterministic 10% slice of the published train set as validation for early stopping (seed=42), yielding 24,295 train, 2,699 val, and 6,749 test. The crisprHAL paper does not publish per-gene annotations for the WT-SpCas9 corpus, so leave-genes-out is not available here; we document this as a methodological limitation.

2.3 Features

For both corpora, we implement a feature extractor in the style of Rule Set 2 [1]: position-specific mononucleotide and dinucleotide one-hot encodings, position-independent mono- and dinucleotide counts, T_m of the full 20-nt protospacer plus three regional T_m windows (positions 1–5, 6–13, 14–20 of the protospacer), GC content of the protospacer, and a binary flag for the presence of “TTTT” (a Pol III transcription terminator that disrupts sgRNA expression). For the mammalian path we additionally include two biological-context features from the Doench corpus: “Percent Peptide” (position of the cut within the coding sequence) and “AA Cut position” (residue index); these are z-scored using training-set statistics persisted alongside the model checkpoint for runtime fill-in. The mammalian feature vector is 610 (sequence) + 2 (bio) = 612; the bacterial feature vector is 870 sequence features over the 43-mer window, with no equivalent bio-context features available in the corpus.

For transformer inputs, each 30-mer (mammalian) or 43-mer (bacterial) is encoded as a sequence of integer nucleotide indices and consumed by an embedding layer.

2.4 Baseline model

For both corpora we train **xgboost** [3] gradient-boosted regressors with similar hyperparameters: max_depth=5–6, learning_rate=0.08–0.10, subsample=0.9, colsample_bytree=0.9, with early stopping

on the validation split (patience 30 rounds). Targets are raw activity scores; xgboost is scale-agnostic and the metrics we report (Spearman, Pearson, AUC) are scale-invariant or threshold-based.

2.5 Transformer model

For both corpora we use an encoder-only transformer with the same default architecture: four pre-LayerNorm encoder layers, $d_{\text{model}}=64$, four heads, $\text{dim}_{\text{feedforward}}=256$, dropout=0.1. The output is a learned embedding mean-pooled across positions, passed through a final LayerNorm and a linear head with a sigmoid activation. For the mammalian path the model additionally fuses two z-scored bio-context features via a small MLP (16 hidden units) concatenated to the pooled sequence embedding before the head; total parameter count is $\approx 204,000$. For the bacterial path the bio-MLP is omitted; total parameter count is $\approx 200,000$.

We train with AdamW ($\text{lr}=3\text{e-}4$, weight decay $1\text{e-}4$), linear warmup over 3–5 epochs followed by cosine decay to zero by 40–60 epochs, batch size 64–128, gradient clipping at max-norm 1.0, MSE loss against the rank-normalised target in $[0, 1]$. Early stopping on validation Spearman with patience 10–15 epochs. Training takes 3–5 minutes on a CPU.

For the bacterial path, we run two variants: **Path B (fresh)** trains all weights from random initialisation, and **Path A (transferred)** initialises the embedding and encoder weights from the mammalian Phase-2 checkpoint, leaving the positional embedding (30→43 positions) and the linear head to train from scratch; the mammalian bio-MLP branch is dropped entirely. Only the tensors whose shapes exactly match across architectures are transferred.

2.6 Window adaptation for the bacterial corpus

The published crisprHAL WT-SpCas9 rows are 378 nt long, with the labeled target's PAM always at positions 209–211. Their CNN+RNN architecture scans the full 378-nt window. To feed this corpus into our 43-mer transformer without changing architecture, we crop each row at the canonical PAM offset, taking sequence[179:222] — yielding a 43-mer composed of a 10-nt 5' flank, the 20-nt protospacer at internal positions 10–29, the 3-nt NGG PAM at internal positions 30–32, and a 10-nt 3' flank. We verified this crop preserves the GG of the PAM at the expected position across all 33,743 rows.

2.7 Evaluation

Test-set predictions from each model are scored against the held-out true activity values using:

Spearman ρ as the primary metric (rank-based, matches the CRISPR literature standard).

Pearson r as a secondary linear-correlation metric.

AUC for binary classification of “score above the test-set median activity,” matching one of the auxiliary metrics reported in Doench 2016.

All three metrics are reported with bootstrap 95% confidence intervals ($n=1,000$ resamples, fixed seed=42 so the same intervals are reproducible between models). Metrics are additionally **stratified by protospacer GC content** into three buckets — low (<0.40), mid ($0.40\text{--}0.70$), and high (≥ 0.70) — to surface architecture-specific performance differences across the GC distribution.

A single shared evaluation module produces all reported numbers, so identical bootstrap resamples are used across models on the same test split.

3. Results

Our xgboost reproduction of Rule Set 2 on the Doench 2016 leave-genes-out test split (550 rows; held-out genes CUL3, HPRT1, NF2, TADA1) achieves:

Metric

Our reproduction

Azimuth V3 on same rows

Paper (Doench 2016)

Spearman ρ

+0.459 [+0.389, +0.527]

+0.665

$\approx$ +0.55

Pearson r

+0.471 [+0.404, +0.531]

+0.678

—

AUC (>median)

+0.670 [+0.616, +0.719]

+0.783

—

The reproduction lands within the bootstrap CI of the paper's reported ~0.55 Spearman. The reproduction-vs-Azimuth Spearman correlation (the rank correlation between our predictions and Microsoft's released V3 model's predictions on the same rows) is +0.825, confirming we are faithfully replicating the Rule Set 2 ranking rather than fitting to noise. Azimuth V3's apparent higher absolute Spearman is partly due to information leakage: its training included rows that appear in our test split.

Trained on the same train+val split with bio-context features concatenated to the pooled sequence embedding, the transformer's test-set performance is:

Metric

Baseline (xgboost + bio)

Transformer (with bio)

Δ

Spearman ρ

+0.459 [+0.389, +0.527]

+0.420 [+0.343, +0.488]

-0.04

Pearson r

+0.471 [+0.404, +0.531]

+0.449 [+0.379, +0.513]

-0.02

AUC (>median)

+0.670 [+0.616, +0.719]

+0.669 [+0.613, +0.720]

± 0.00

CI's heavily overlap on every metric, and AUC is identical to three decimal places. Under any reasonable threshold for a “significant” difference (CI exclusion of zero), the architectures are statistically indistinguishable. The transformer’s reproduction-vs-Azimuth Spearman is +0.731, lower than the baseline’s +0.825 but still well above zero, indicating it learns much of the same ranking pattern.

Stratified by GC bucket, the architectures show different failure modes: the transformer slightly outperforms the baseline on AT-rich gRNAs (low-GC Spearman 0.535 vs 0.519), matches in the bulk middle, and underperforms substantially on the small high-GC bucket (n=36, Spearman 0.164 vs 0.552). The high-GC bucket’s CI is wide and includes negative values; we do not claim a robust GC-dependent transformer weakness from a single 36-row slice.

3.3 The xgboost baseline clearly beats the transformer on bacterial

On the crisprHAL WT-SpCas9 held-out test set (6,749 rows), the three-way comparison is:

Model

Spearman ρ

Pearson r

AUC (>median)

xgboost baseline (870 features)

+0.626 [+0.608, +0.642]

+0.630 [+0.612, +0.647]

+0.826 [+0.816, +0.835]

Transformer, Path B (fresh)

+0.544 [+0.526, +0.562]

+0.546 [+0.529, +0.565]

+0.787 [+0.777, +0.798]

Transformer, Path A (Phase-2 → bacterial)

+0.530 [+0.512, +0.548]

+0.533 [+0.514, +0.551]

+0.780 [+0.769, +0.790]

Bootstrap CIs do not overlap between xgboost and either transformer variant on any metric. The performance gap of approximately 0.08 Spearman is statistically significant. Both transformer variants substantially outperform a randomly-ranking baseline (Spearman 0) but cleanly lose to gradient-boosted trees over Rule Set 2-style features.

3.4 Mammalian → bacterial transfer learning does not help

Path A initialises the transformer’s embedding, encoder, and final LayerNorm from the Phase-2 mammalian checkpoint; only the positional embedding (size mismatch 30→43) and output head are trained from scratch. Despite this warm start, Path A’s test Spearman of +0.530 [0.512, 0.548] is marginally **lower** than Path B’s +0.544 [0.526, 0.562] — within bootstrap CI overlap, so we do not claim Path A is significantly worse, but the mammalian pretraining clearly provides no measurable benefit. The encoder representations learned on Doench 2016 (mammalian cell screens with chromatin and expression-state context) do not transfer usefully to bacterial CRISPR-Cas9 activity in plasmid contexts.

GC bucket

n

xgboost

Path B

Path A

low (<0.40)

985

+0.552

+0.475

+0.462

mid (0.40–0.70)

5,416

+0.625

+0.544

+0.528

high (≥ 0.70)

348

+0.711

+0.555

+0.547

Notable: on the bacterial corpus, xgboost performs **best in the high-GC bucket** (Spearman 0.711) — the opposite of the pattern observed in the mammalian transformer, where high-GC was the weakest slice. The transformer variants show flat performance across buckets with no high-GC strength.

4. Discussion

The central finding is the **contrast in head-to-head outcomes across the two corpora**: a 200-k-parameter encoder-only transformer matches a hand-engineered gradient-boosted regressor on Doench 2016 (5,310 mammalian sgRNAs) but cleanly loses on the crisprHAL WT-SpCas9 corpus (33,743 *E. coli* sgRNAs), despite the bacterial corpus being substantially larger. Three plausible drivers, ordered by what we believe is their likely contribution:

(i) Cleaner bacterial signal allows hand-engineered features to dominate. The bacterial corpus measures cleavage in a controlled plasmid system designed by Ham et al. to separate activity from toxicity. Mammalian CRISPR screens, by contrast, fold in chromatin accessibility, expression state, nucleosome positioning, and other context-dependent factors that do not appear in plasmid assays. Rule Set 2-style features (position-specific mono- and dinucleotide one-hots, regional Tms, GC, TTTT) capture the local-sequence determinants of Cas9 cleavage chemistry that dominate when those mammalian confounders are absent. Where the mammalian corpus has substantial variance the engineered features do not see — variance a flexible transformer can in principle pick up — the bacterial corpus has comparatively little. Boosted trees with strong features are very hard to beat in that regime.

(ii) The 43-mer crop discards bacterial signal that crisprHAL preserves. Ham et al. designed crisprHAL's CNN+RNN around the full 378-nt context window, plausibly because long-range flanking sequence

carries information about Cas9 binding and cleavage (e.g., DNA bending propensity, accessibility motifs, secondary-structure energetics). To feed the bacterial corpus into our 43-mer architecture we crop to a 10 + 20 + 3 + 10 window centred on the labeled PAM. This discards 335 nt per row, and the architecture choice the crisprHAL authors made is itself evidence that the discarded sequence carries predictive signal.

(iii) The transformer architecture and training recipe are not well-matched to a clean 24-k-row corpus.

The Phase-2 architecture was selected to be plausible for the small (≈ 4 k row) mammalian training set. With ≈ 24 k bacterial training rows we could afford a substantially larger model — but a 200-k-parameter transformer is in a regime where the inductive bias of gradient-boosted trees over good features is structurally hard to outdo. crisprHAL’s published architecture (CNN+RNN, not transformer) reflects an architectural choice made by domain experts after, we infer, evaluating multiple options.

The failure of cross-kingdom transfer (Path A) is consistent with mechanism (i): if mammalian and bacterial Cas9 cleavage depend on distinct factors (chromatin context in mammals vs. plasmid-local sequence in bacteria), then the encoder representations the model learned for mammalian targets are not useful — and may be marginally counterproductive — for bacterial targets. The slightly lower central estimate for Path A (0.530 vs. 0.544 for Path B), although within CI overlap, is consistent with mild negative transfer.

The bacterial transformer’s flat performance across GC buckets, contrasted with xgboost’s increase from 0.55 (low) to 0.71 (high), is a separately interesting observation. We interpret it as the engineered GC-related features (GC content of the protospacer, regional Tms, and the position-specific dinucleotide one-hots) directly encoding what the transformer would otherwise have to learn from raw nucleotide context — with the engineered features being the more sample-efficient route at this corpus size.

We emphasise three methodological points: (1) **no test-set iteration was performed in either phase**; the reported numbers are the result of a single training run with locked hyperparameters in each case. (2) The mammalian and bacterial test sets were held out separately and never touched during architecture or feature-set decisions on the other corpus. (3) The Phase-3 transformer checkpoints (both fresh and transferred) remain in the project repository for reproducibility but were not exposed in the deployed designer because the xgboost baseline is the better model on the corpus we have.

5. Limitations

Single seed. Each training run uses seed=42 throughout. The bootstrap CIs on the reported metrics quantify within-run uncertainty against the test set, but variance from re-initialisation is not characterised. A multi-seed ensemble is the natural next step.

No hyperparameter sweep. Architecture defaults (4 layers, d_model=64, 4 heads, dim_feedforward=256) were locked across both corpora. A sweep on the bacterial corpus, where we have substantially more data, might close the central-estimate gap to baseline; we have not attempted one.

Window cropping on the bacterial corpus. We crop crisprHAL's 378-nt rows to 43 nt at the canonical PAM offset, which discards 335 nt of context per row. The published crisprHAL architecture (CNN+RNN over 378-mer) was designed for long context. A long-context transformer or a CNN+RNN replication would be the appropriate test of the cropping-loss hypothesis.

Bio-context features asymmetry. The mammalian corpus carries two biological-context features (Percent Peptide, AA Cut position) that contributed approximately 0.05 Spearman in our internal ablation. The bacterial corpus has no equivalent. Like-for-like architecture comparisons across corpora are therefore not strictly possible.

No off-target prediction. The reported models predict on-target activity only; off-target risk assessment (the second major axis of practical CRISPR design) is out of scope for this report.

Leave-genes-out is not feasible for bacterial. The Doench 2016 corpus carries explicit gene labels; the crisprHAL WT-SpCas9 corpus does not. We use leave-genes-out for the mammalian evaluation (the same protocol the Doench paper uses) but a random 10% validation slice on the bacterial corpus. Whether the published bacterial train and test sets share fine-scale structure that artificially inflates either model's test performance is something we cannot directly check.

6. Future work

A small, well-defined set of follow-ups would close the most important gaps:

1. **Multi-seed evaluation.** Three or five training runs with different seeds; report mean $\pm$ std as well as bootstrap CIs. Cheap to execute (each run takes 3–5 minutes on CPU); resolves the within-architecture variance question.
2. **Long-context bacterial transformer.** Train a transformer that accepts the full 378-nt crisprHAL window, with a larger parameter count to match the increased input dimensionality. Directly tests the cropping-loss hypothesis (mechanism ii above).
3. **CNN+RNN architecture as the bacterial baseline.** A faithful reimplementation of crisprHAL's published architecture trained on the same train/val/test partition as our work would let us compare (a) our xgboost numbers against the published CNN+RNN and (b) our small transformer against a published architecture choice that domain experts converged on. The crisprHAL repository ships pre-trained weights and the architecture is documented in the supplementary materials of Ham et al. 2023.
4. ***Bacillus subtilis* extension.** *B. subtilis* is the dominant industrial production strain for enzymes, surfactins, and small-molecule biosynthesis; a working bacterial CRISPR designer that targets it would have direct commercial relevance. No Doench-equivalent *B. subtilis* corpus appears to exist publicly at the time of writing; possible paths are fine-tuning on smaller published *B. subtilis* CRISPR studies (Zou et al. 2022, Kim et al. 2025) or curating a *B. subtilis* corpus de novo.

5. **Off-target prediction.** Integrating an exact off-target search (Cas-OFFinder, Bae et al. 2014) and an off-target scoring function (CFD, Doench 2016) would complete the designer as a usable tool rather than a benchmark exercise.
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7. Conclusion

We compared a small sequence-only transformer head-to-head against a hand-engineered gradient-boosted baseline for CRISPR-Cas9 on-target activity prediction across two corpora. The transformer ties the baseline on the mammalian Doench 2016 corpus and loses cleanly on the bacterial crisprHAL WT-SpCas9 corpus. Mammalian→bacterial transfer learning did not help. The contrast is consistent with hand-engineered Rule Set 2 features capturing most of the available bacterial signal (mechanism i), the long-context architecture choice in crisprHAL’s published design preserving information our 43-mer crop discards (mechanism ii), and the small-transformer-on-clean-data regime being structurally favourable to gradient-boosted trees (mechanism iii). The result is a useful data point for practitioners deciding whether to reach for a deep architecture or a strong classical baseline at CRISPR-Cas9 corpus scales: the answer depends on the corpus, and “transformer wins” is not the default conclusion.

A working multi-organism CRISPR designer (FastAPI backend + React frontend) using the better model per organism (Rule Set 2 + Phase-2 transformer for mammalian; xgboost baseline for bacterial) accompanies this report as a working artifact.

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Appendix A — Reproducibility

All training and evaluation code, training-set splits, trained model checkpoints (for both corpora), and the deployed designer (FastAPI + React) are available in the project repository. Reproduction:

```
make data      # Doench 2016 corpus via the Azimuth repository
make validate  # schema and integrity assertions on the corpus
make splits    # deterministic leave-genes-out + LPT bin-packing
make baseline  # train + eval Rule Set 2 reproduction

python ml/scripts/train_ours.py      # mammalian transformer
python ml/scripts/eval_ours.py       # mammalian eval

python ml/scripts/download_data_bacterial.py
python ml/scripts/validate_data_bacterial.py
python ml/scripts/make_splits_bacterial.py
python ml/scripts/train_baseline_bacterial.py
python ml/scripts/eval_baseline_bacterial.py
python ml/scripts/train_ours_bacterial.py      # Path B
python ml/scripts/train_ours_bacterial.py --transfer-from ml/checkpoints/ours_best.pt # Path A
python ml/scripts/eval_ours_bacterial.py # produces docs/bacterial_results.md
```

All random seeds are fixed (seed=42). All metrics use a shared ml/eval_common.py module with a fixed bootstrap seed for reproducible CIs.

End of report.